

TEST YOUR KNOWLEDGE

1. Ten coins are tossed simultaneously. Find the probability of getting at least seven heads.
2. The probability of any ship of a company being destroyed on a certain voyage is 0.02. The company owns 6 ships for the voyage. What is the probability of :
 - (i) losing one ship
 - (ii) losing at most two ships
 - (iii) losing none.
3. The probability that a man aged 60 will live to be 70 is 0.65. What is the probability that out of ten men now 60, at least 7 would live to be 70 ?
4. The incidence of occupational disease in an industry is such that the workers have a 20% chance of suffering from it. What is the probability that out of six workers chosen at random, four or more will suffer from the disease ?
5. The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that
 - (i) exactly two will be defective
 - (ii) at least two will be defective
 - (iii) none will be defective.
6. If the chance that one of the ten telephone lines is busy at an instant is 0.2.
 - (i) What is the chance that 5 of the lines are busy ?
 - (ii) What is the probability that all the lines are busy ?
7. If on an average 1 vessel in every 10 is wrecked, find the probability that out of 5 vessels expected to arrive, at least 4 will arrive safely.
8. A product is 0.5% defective and is packed in cartons of 100. What percentage contains not more than 3 defectives ?
9. A bag contains 5 white, 7 red and 8 black balls. If four balls are drawn, one by one, with replacement, what is the probability that
 - (i) none is white
 - (ii) all are white
 - (iii) at least one is white
 - (iv) only 2 are white ?
10. In a hurdle race, a player has to cross 10 hurdles. The probability that he will clear each hurdle is $\frac{5}{6}$. What is the probability that he will knock down fewer than 2 hurdles ?
11. Fit a binomial distribution for the following data and compare the theoretical frequencies with the actual ones :

$x :$	0	1	2	3	4	5
$f :$	2	14	20	34	22	8

12. If the sum of mean and variance of a binomial distribution is 4.8 for five trials, find the distribution.
13. If the mean of a binomial distribution is 3 and the variance is $\frac{3}{2}$, find the probability of obtaining at least 4 success.
14. In 800 families with 5 children each, how many families would be expected to have (i) 3 boys and 2 girls, (ii) 2 boys and 3 girls, (iii) no girl (iv) at the most two girls. (Assume probabilities for boys and girls to be equal.)
15. In 100 sets of ten tosses of an unbiased coin, in how many cases do you expect to get
(i) 7 heads and 3 tails (ii) at least 7 heads ?
16. The following data are the number of seeds germinating out of 10 on damp filter for 80 sets of seeds. Fit a binomial distribution to this data :
- | | | | | | | | | | | | | |
|-------|---|----|----|----|---|---|---|---|---|---|----|-------|
| $x :$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Total |
| $f :$ | 6 | 20 | 28 | 12 | 8 | 6 | 0 | 0 | 0 | 0 | 0 | 80 |
- [Hint. Here $n = 10$, $N = 80$, Mean $= \frac{\sum fx}{\sum f} = 2.175 \therefore np = 2.175$ etc.]
17. A bag contains 10 balls each marked with one of the digits 0 to 9. If four balls are drawn successively with replacement from the bag, what is the probability that none is marked with the digit 0 ?
18. A box contains 100 tickets each bearing one of the numbers from 1 to 100. If 5 tickets are drawn successively with replacement from the box, find the probability that all the tickets bear numbers divisible by 10.
19. The probability that a bomb dropped from a plane will strike the target is $\frac{1}{5}$. If six bombs are dropped, find the probability that (i) exactly two will strike the target, (ii) at least two will strike the target.
20. In sampling a large number of parts manufactured by a machine, the mean number of defectives in a sample of 20 is 2. Out of 1000 such samples, how many would be expected to contain at least 3 defective parts.

Answers

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| 1. $\frac{11}{64}$ | 2. (i) 0.1085, (ii) 0.9997, (iii) 0.8858 | 3. 0.514 |
| 4. $\frac{53}{3125}$ | 5. (i) 0.2301 (ii) 0.3412 (iii) 0.2833 | |
| 6. (i) 0.02579 (ii) 1.024×10^{-7} | 7. 0.91854 | 8. 99.83 |
| 9. (i) $\frac{81}{256}$ (ii) $\frac{1}{256}$ (iii) $\frac{175}{256}$ (iv) $\frac{27}{128}$ | 10. $\frac{5}{2} \left(\frac{5}{6}\right)^9$ | |
| 11. $100 (0.432 + 0.568)^5$ | 12. $\left(\frac{1}{5} + \frac{4}{5}\right)^5$ | 13. $\frac{11}{32}$ |
| 14. (i) 250, (ii) 250, (iii) 25, (iv) 400 | 15. (i) 12 nearly (ii) 17 nearly | |
| 16. $80 (0.7825 + 0.2175)^{10}$ | 17. $\left(\frac{9}{10}\right)^4$ | 18. 0.00001 |
| 19. (i) 0.246 (ii) 0.345 | 20. 323. | |